



SQL Server 2025 and Azure SQL: AI where business data lives

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Slide and demo: bit.ly/1nn0vAI-2026-sgovoni



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Agenda

- Introduction to AI-ready evolution of SQL Server 2025 and Azure SQL
- Using vector data types and perform semantic search
 - Preparing SQL instance
 - Preparing data for semantic search
 - Generating embeddings with Azure OpenAI and T-SQL
 - Perform semantic search with VECTOR_DISTANCE, VECTOR_SEARCH
- Vector indexes in Azure SQL Database



Introduction to AI-ready evolution

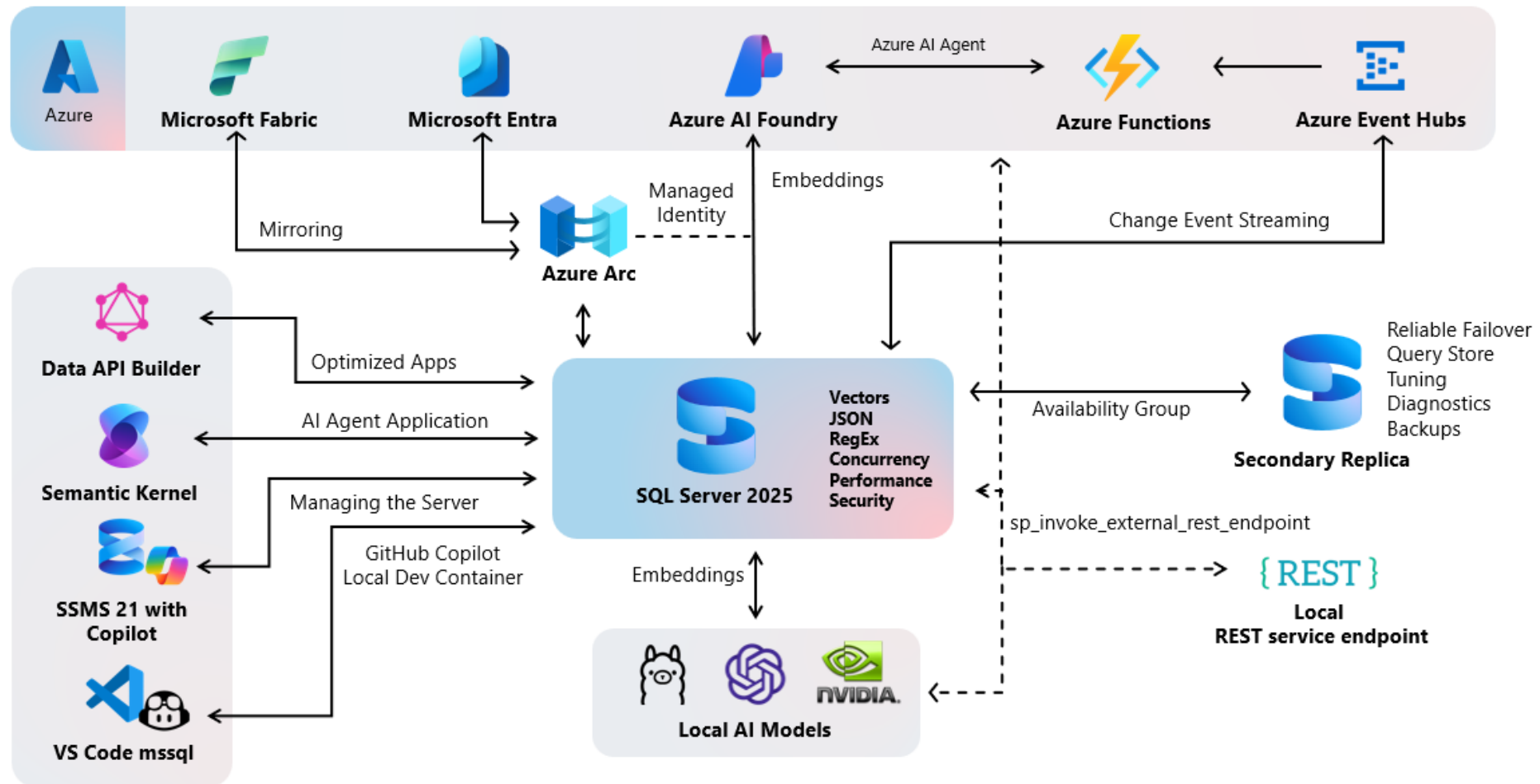
In SQL Server 2025 and Azure SQL



SQL Server 2025 AI capabilities

- SQL Server 2025 introduces AI capabilities directly into the database engine
 - Vector data type
 - Connectivity with Azure OpenAI and other frameworks
 - Embedding generation
 - Vector search and vector indexes
 - sys.sp_invoke_external_rest_endpoint (available from SQL Server 2022)
 - Foundry models sold by Azure

SQL Server 2025 AI capabilities





Vector data types and semantic search

From zero to AI



Preparing SQL instance

- The sys.sp_invoke_external_rest_endpoint system stored procedure is available in SQL Server 2025 and Azure SQL Managed Instance, but it's disabled by default
 - It's enabled by default in Azure SQL Database and in SQL Fabric
- To call external REST endpoint, you must enable the external rest endpoint enabled server configuration option using sp_configure

Preparing data for semantic search

- Raw data is usually optimized for relational queries, filters, and joins
- Semantic search needs a richer text built from relevant content and metadata
- A dedicated table helps prepare and control the text used to generate embeddings
- This improves search quality by making data searchable by meaning, not only by keywords

Generating embeddings with Azure OpenAI and T-SQL

- Use an EXTERNAL MODEL to connect SQL Server to the embedding model endpoint
- Generate embeddings directly from T-SQL with AI GENERATE EMBEDDINGS
- Store the vector representation next to the prepared text in the database
- Process data in batches to control execution time, throughput, and cost

Introduction to semantic search

- Text can be transformed into embeddings, which are vectors that represent meaning
- Similar concepts produce vectors that are close to each other
- A natural language question is converted into an embedding that represents its meaning
- The embedding of the question is compared with stored embeddings to find semantically similar content

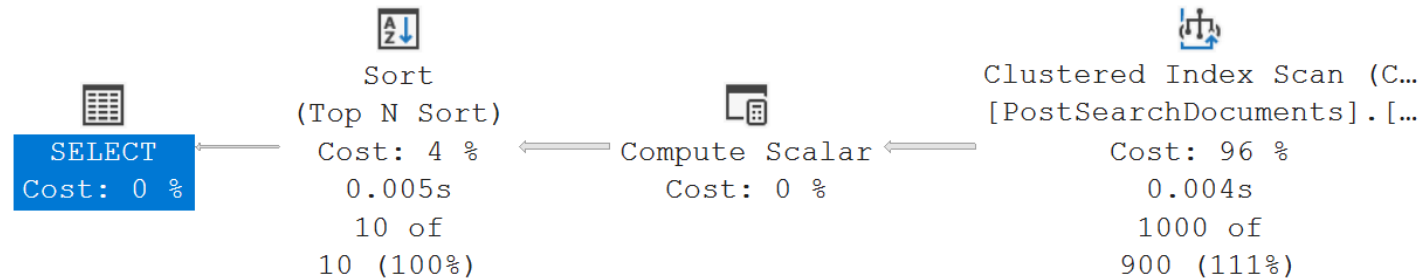
Introduction to semantic search

- Semantic search can be combined with traditional SQL filters and metadata
- Results are ranked by distance: lower distance means higher semantic similarity
- Execution plans are key to understanding how semantic search queries behave
- Performance depends on whether the query uses exact distance calculation or indexed approximate search

Semantic search with VECTOR_DISTANCE

- VECTOR_DISTANCE is designed for exact similarity checks, not approximate retrieval
- Every candidate embedding must be evaluated against the embedding of the “question”
- Existing ANN vector indexes do not change this behavior
- Best suited for scenarios where exact results matter and the candidate set is controlled

Semantic search with VECTOR_DISTANCE



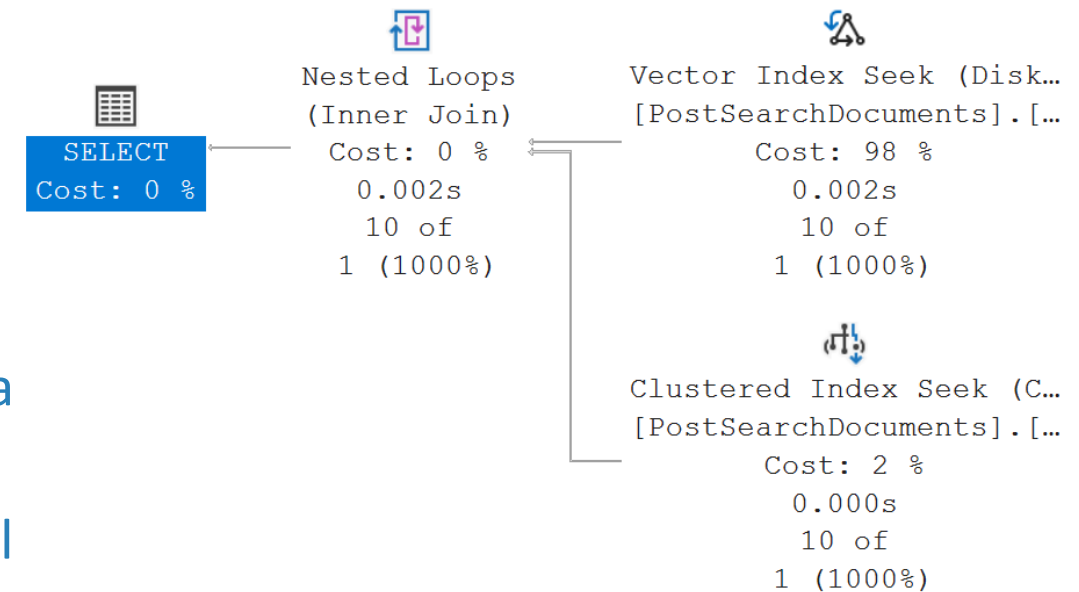
- The query optimizer scans the candidate rows instead of performing an index seek
- VECTOR_DISTANCE is evaluated for each candidate embedding
- A Compute Scalar operator performs the distance calculation
- Sorting is needed to return the closest matches first
- This behavior is transparent and predictable, but less scalable on large candidate sets

Vector indexes

- A vector index
 - Helps SQL Server search for similar embeddings more efficiently
 - Is built to support ANN (approximate nearest-neighbor) search
- SQL Server uses an algorithm known as DiskANN to navigate similar embeddings more efficiently
- The goal is to reduce the cost of scanning large candidate sets
- Performance improves by accepting a controlled approximation

Semantic Search with VECTOR_SEARCH

- To use the vector index, the query must follow specific requirements
 - The use of VECTOR_SEARCH
 - VECTOR_SEARCH must be used with TOP (N) WITH APPROXIMATE
 - Results must be ordered by the distance column
- When the requirements are met, the execution plan can use vector index seek
- Without approximate search, the optimizer can fall back to a scan



Important considerations and limitations

- Vector index features are still in preview in SQL Server 2025
- In SQL Server 2025, creating a vector index currently makes the table read-only
 - INSERT, UPDATE, DELETE, and MERGE operations are not supported on that table after index creation
- Relational filters may be applied after vector search, depending on the index version

Important considerations and limitations

- In vector search, SQL Server may fetch more vectors than requested before applying final ranking
 - This over-fetching helps reduce the approximation error
- Always validate syntax, limitations, and execution plans in your environment



Vector indexes in Azure SQL Database

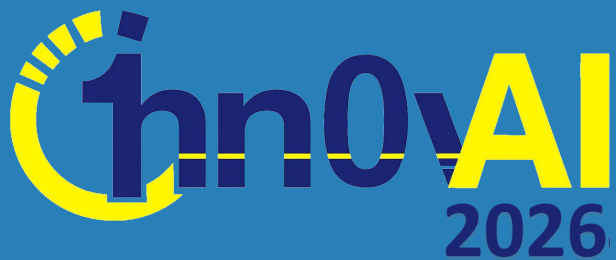


Vector Indexes: SQL Server 2025 vs Azure SQL

- SQL Server 2025 currently uses the first vector index implementation
- Azure SQL Database uses the latest vector index format, version 3
- The index version can be checked through [sys.vector_indexes](#)
- The same CREATE VECTOR INDEX syntax can lead to different behaviors depending on the platform

Vector Indexes: DML Support

- In SQL Server 2025, creating a vector index currently makes the table read-only
 - INSERT, UPDATE, DELETE, and MERGE fail after the vector index is created
- In Azure SQL Database, vector-indexed tables remain read-write
 - Azure SQL Database supports DML with automatic vector index maintenance
 - Azure SQL Database also supports TOP (N) WITH APPROXIMATE and iterative relational filtering



Demo

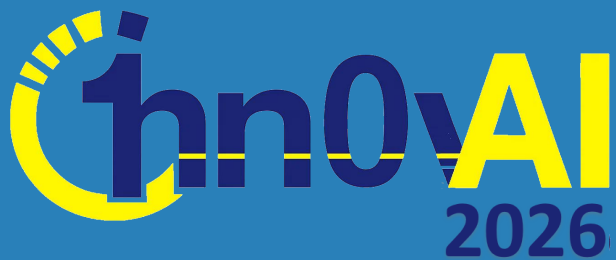


Summary

- Embeddings stored close to relational data enable semantic search directly with T-SQL
- VECTOR_DISTANCE performs exact search: every candidate is evaluated
- VECTOR_SEARCH with a vector index performs approximate nearest-neighbor search
- SQL Server 2025 still has preview limitations, including read-only tables after CREATE VECTOR INDEX
- Azure SQL Database uses vector index version 3, keeps tables read-write, and supports iterative filtering

Resources

- [VECTOR DISTANCE](#)
- [VECTOR SEARCH](#)
- AI where business data lives
 - [Part 1: Preparing Data for Semantic Search](#)
 - [Part 2: Generating Embeddings with Azure OpenAI and T-SQL](#)
 - [Part 3: Semantic Search with T-SQL](#)
 - [Part 4: Indexed Vector Search with SQL Server 2025](#)
 - [Part 5: VECTOR DISTANCE vs VECTOR SEARCH: How Does the Execution Plan Change?](#) (thanks to Riccardo Perico)
 - Part 6: coming soon 😊



Thanks!

